

Automated Seat Belt Detection Using Surveillance Cameras for Indian Traffic: A Deep Learning Approach

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Abstract—The increasing number of road accidents in India highlights the urgent need for effective traffic safety enforcement. Seat belts are essential for minimizing injuries during vehicle collisions, yet compliance remains inconsistent. This paper proposes an automated deep learning based approach for detecting seat belt violations in Indian traffic scenarios. The method leverages lightweight variants of the YOLOv8 object detection model to identify vehicles, determine their direction of movement, and analyze windshield regions to assess seat belt usage. To enhance performance under poor lighting and challenging visibility conditions during the day, Contrast Limited Adaptive Histogram Equalization is applied as a image processing step. Experimental evaluations demonstrate that the proposed system achieves an average precision of 0.875 in detecting seat belt violations. The inclusion of image enhancement improves overall accuracy by 0.57 percentage points. With a processing time of 54.9 milliseconds per frame for completing the detections and classifications, the system is well-suited for real-time deployment in urban environments. This study contributes to the development of intelligent transportation systems by enabling safer and more efficient road monitoring and law enforcement.

Index Terms—Seat Belt Detection, YOLOv8, Deep Learning, Traffic Safety, CLAHE.

I. INTRODUCTION

Road safety remains a critical concern worldwide, with seat belt usage being one of the most effective measures to reduce fatalities in traffic accidents. Despite stringent regulations, a significant number of drivers and passengers neglect seat belts usage, increasing the risk of severe injuries and fatalities. Manual enforcement of seat belt laws is labour-intensive, prone to human error, and inefficient for large-scale monitoring. Consequently, automated seat belt detection systems leveraging deep learning and computer vision have gained significant attention as a means to enhance compliance and support traffic law enforcement.

However, existing seat belt detection systems face several challenges, including low detection accuracy under poor lighting conditions, occlusions caused by vehicle interiors, and variations in seatbelt appearance due to different clothing and seat colours. To address these challenges, this paper

presents a three-stage automated approach for detecting seat belt violations using an optimized YOLOv8-based model and Contrast Limited Adaptive Histogram Equalization (CLAHE) for image enhancement.

The key contributions of this work include:

- A lightweight YOLOv8-based detection pipeline that performs real-time vehicle detection, windshield segmentation, and seat belt classification.
- The application of CLAHE to enhance image quality and improve detection accuracy in low-light conditions.
- A comprehensive evaluation of the proposed system on a diverse dataset, demonstrating high precision and recall in seat belt detection.

By leveraging an optimized deep learning framework, the proposed method enhances detection accuracy, ensuring robust real-time enforcement of seat belt laws.

II. RELATED WORKS

Several deep learning-based approaches have been proposed for seat belt detection, each demonstrating varying levels of accuracy, computational efficiency, and real-time applicability. However, most prior models struggle with key challenges such as poor lighting conditions, occlusions, and variations in seat belt appearance due to different clothing and seat colours.

A Gated Bi-LSTM model with part-to-whole attention achieved 72.3% accuracy but required extensive hyperparameter tuning, making it computationally expensive and less viable for real-time applications [1]. Similarly, a pre-trained Xception model demonstrated high accuracy (99.42%) but had slower inference times, limiting its deployment in traffic surveillance systems [2].

Traditional CNN-based models such as ResNet50, VGG16, and InceptionV3 were tested for feature extraction, with ResNet50 and VGG16 achieving 72% accuracy and InceptionV3 has 56% accuracy. However, these models exhibited lower performance in real-world traffic conditions with complex backgrounds [3]. The YOLOv8 family has shown promise

in real-time object detection tasks, with YOLOv8m achieving 96% accuracy in windshield detection and YOLOv8l-cls obtaining 88.46% accuracy for seat belt classification and 89% accuracy for passenger classification [4]. But with this method, limited contextual information may lead to misinterpretations since it directly detect the windshields of vehicles without detecting the vehicles that are suitable for seat belt detection. Meanwhile, MobileNetV2 attained 89% accuracy, though its performance was highly dependent on dataset quality, limiting its generalisability across diverse traffic conditions [5].

Hybrid approaches, including YOLOv5, ResNet34 with Spatial Pyramid Pooling (SPP) and Power Mean Transformation (PMT), achieved a high seat belt detection accuracy of 98.9%. However, such architectures require significantly more computational power, making them impractical for real-time processing in traffic surveillance systems [6]. Additionally, YOLOv5 for windshield detection and AlexNet for seat belt classification is used in [10]. After detecting the windshield, the hough transformation technique is applied on the windshield to detect its boundaries more precisely for further seatbelt classification. Windshield detection is performed with 93.3% accuracy and classification has obtained 87% accuracy, though the mandatory image processing step increases the overall processing time and makes the model less practical for real-time implementations. Lightweight models like Tiny YOLO (93% accuracy) were computationally efficient but suffered from classification instability in dynamic real-world conditions [7].

Another promising approach, NADS-Net, developed for in vehicle monitoring, achieved 89% accuracy in seat belt detection with ResNet50 with Feature Pyramid Network (FPN). But adding FPN increases the computational load, results in higher memory consumption. [9]. Lastly, a hybrid CNN-SVM model reached an 85% Correct Identification Rate (CIR) by leveraging CNN for initial detection and SVM for post-processing refinement [8].

These studies highlight the trade-offs between accuracy, computational efficiency, and real-time applicability. The proposed approach builds upon these insights, utilising YOLOv8-based models with CLAHE image enhancement to improve seat belt detection under challenging real-world conditions.

III. METHODOLOGY

A. Dataset Details

A custom dataset was created using surveillance cameras with minimum 5MP ZTP (Zoom, Tilt and Pan) for overall surveillance. From the collected real-time videos, a total of 2,000 frames were extracted which covering occlusions, variable lighting, heavy traffic, and partial obstructions.

Annotations on these frames were performed using Label Studio for five vehicle classes: two-wheeler, three-wheeler, four-wheeler, bus, and truck. Windshield detection and passenger classification annotations were also included. The data set was divided into an 80:10:10 ratio for training, validation and testing.

B. Proposed Method

The proposed system consists of a multi-stage deep learning pipeline, as illustrated in Fig.1 and Fig.2.

Hyper parameter tuning of YOLOv8 models are performed on the vehicle detection validation dataset, for object detection, and best result obtained for YOLOv8n and YOLOv8s, with 50 epochs, 32 batch size, Adam optimizer and a learning rate of 0.001111, considering both accuracy and processing time. The comparison table showing the performance of each model is given in Table I. Since lightweight models of YOLOv8 provide better trade-off between the speed and accuracy in object detection, these models along with other lightweight models of YOLO are trained in the training dataset for vehicle detection and the result is shown in Table II.

Hyper parameter tuning for object classification is performed on the vehicle classification validation dataset and best result achieved with the YOLOv8n-cls model with 21 epochs, 16 batch size and 0.001667 learning rate. Hence, all the object detection and object classification tasks are performed with YOLOv8n and YOLOv8n-cls models respectively with these optimal parameter values using NVIDIA T4 GPU.

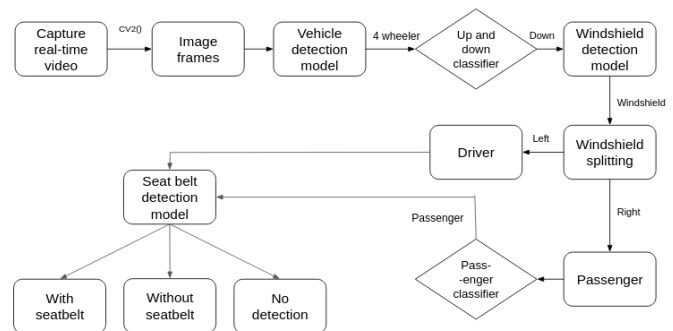


Fig. 1: Proposed System for Seatbelt Detection

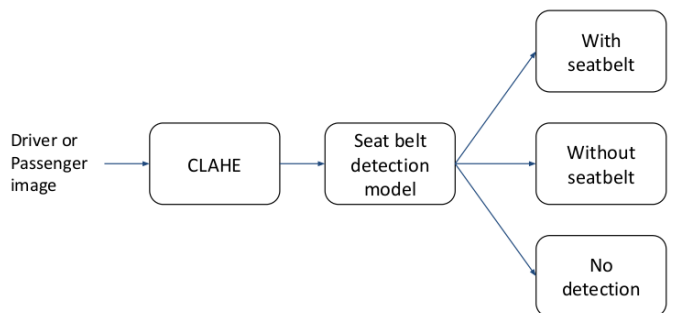


Fig. 2: Case of Without Seatbelt or No Detection Predictions

The proposed system follows a multi-stage deep learning pipeline for seat belt violation detection. It begins with vehicle detection, where YOLOv8n model identify four wheelers from traffic footage. A movement classification model, YOLOv8n-cls, then determines whether the vehicle is moving towards or away from the camera, ensuring seat belt analysis is

TABLE I: Performance comparison of YOLOv8 models for object detection hyperparameter tuning

Models	P	R	mAP50	mAP50-95	Processing time(ms)	Inference time(ms)
Yolov8n	0.927	0.88	0.952	0.776	2.1	1.9
Yolov8s	0.93	0.887	0.96	0.778	3.5	3.3
Yolov8m	0.90	0.896	0.974	0.776	2.0	5.4
Yolov8l	0.938	0.90	0.94	0.778	1.2	9.1
Yolov8x	0.932	0.90	0.943	0.779	1.1	16.1

TABLE II: Performance Comparison of YOLO Models for Vehicle Detection

Model	TIT (ms)	P	R	mAP50	mAP50-95
YOLOv8n	13.9	0.85	0.866	0.923	0.736
YOLOv8s	17.0	0.862	0.862	0.920	0.734
YOLOv9t	25.6	0.843	0.883	0.928	0.736
YOLOv9s	33.7	0.817	0.902	0.925	0.745
YOLOv10n	15.8	0.843	0.866	0.910	0.719
YOLOv10s	18.4	0.861	0.862	0.920	0.736
YOLOv11n	17.3	0.826	0.895	0.924	0.736
YOLOv11s	17.3	0.849	0.883	0.926	0.738

conducted only on vehicles with a clearly visible windshield and occupants.

Next, a windshield detection model isolates the driver and passenger sections for further analysis. If a passenger is detected, a seat belt classification model evaluates the presence of a seat belt, categorising images as “with seat belt” or “without seat belt”. For cases with low confidence scores or poor contrast between the seat belt and garments, CLAHE enhancement is applied to improve visibility.

Since the proposed model designed for real-time processing, it balances accuracy and computational efficiency using lightweight YOLOv8 models for detection and classification, making it suitable for large-scale traffic surveillance and enforcement applications.

C. Evaluation Metrics

Standard object detection metrics were used to assess model performance:

- Precision (P): Measures the proportion of correctly detected seat belts among all detections. It is given by [4]:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

where TP is True Positives, and FP is False Positives. High precision indicates a low false-positive rate.

- Recall (R): Indicates how many actual seat belt instances are correctly detected. It is calculated as [4]:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

where FN is False Negatives. A high recall ensures that most seat belt violations are identified.

- F1 score: The F1 score is the harmonic mean of precision and recall, providing a balanced measure of a model’s accuracy, especially in imbalanced datasets.
- Mean Average Precision (mAP@50): Measures detection performance by computing the average precision (AP) at an Intersection over Union (IoU) threshold of 0.50 [4]. It evaluates how well the model differentiates between seat belt and non-seat belt instances.
- Mean Average Precision (mAP@50-95): Assesses the model’s performance across multiple IoU thresholds (0.50 to 0.95, in 0.05 increments), providing a more comprehensive measure of detection robustness across varying degrees of object overlap.

These metrics collectively evaluate the accuracy, reliability, and effectiveness of the seat belt detection framework, ensuring its applicability for real-world deployment in traffic monitoring systems.

IV. RESULTS AND DISCUSSIONS

The proposed automated seat belt detection system was evaluated on a custom dataset using various YOLOv8 models. The system’s effectiveness was assessed based on precision, recall, F1-score, and inference time, ensuring a comprehensive evaluation of both accuracy and computational efficiency.

- Vehicle Detection

According to Table II(where TIT implies Total Inference Time), YOLOv8n is the better model of vehicle detection in real-time as it takes only 13.9 ms for processing. The model achieved 85% precision and 86.6% recall, demonstrating robust vehicle detection capabilities in diverse traffic conditions. Additionally, YOLOv8n maintained a high precision of 94.9% for four-wheeler detection, ensuring reliable classification.

An example output from the YOLOv8n model detecting four-wheeler vehicles in traffic footage is shown in Fig. 3.



Fig. 3: Detected Four Wheeler Vehicles

- Vehicle Movement Classification

After detecting vehicles, YOLOv8n-cls was used to classify their movement direction. Vehicles moving towards the camera are detected as “Down”, and those moving away from the camera as “Up”. Performance of the model for classification is given in Table III which demonstrate the robustness in classifying vehicle motion.

The confusion matrix showing the classification performance of YOLOv8n-cls in categorizing vehicle movement direction (up/down) is given in Fig. 4.

Vehicle movement classification results on the detected four wheeler vehicles are illustrated in Fig. 5.

TABLE III: Vehicle Classification Performance of YOLOv8n-cls Model

Classes	top1_acc	top5_acc
all	0.98	1.00

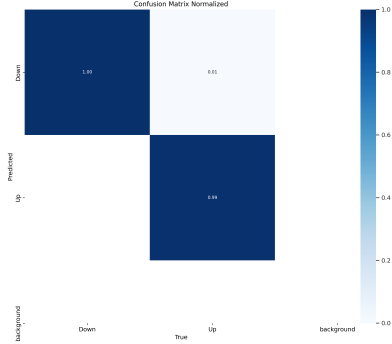


Fig. 4: Vehicle Classification - Normalized Confusion Matrix

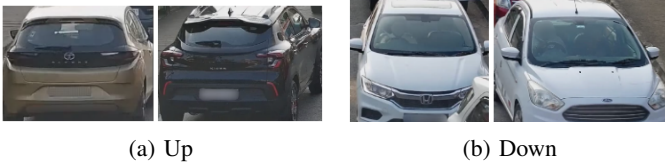


Fig. 5: Vehicle Movement Classification Results

• Windshield Detection and Passenger Classification

To ensure accurate seat belt detection, windshield regions were extracted from detected vehicles. The YOLOv8n model achieved 99.7% precision, ensuring high-quality segmentation, and is shown in Table IV.

TABLE IV: Windshield Detection Performance of YOLOv8n Model

Classes	P	R	mAP@50	mAP@50-95
all	0.997	1.00	0.995	0.841

The confusion matrix [Fig. 6], Precision-Recall curve, and F1 score curve [Fig. 7] represent the performance of the YOLOv8n model in detecting windshields. Fig. 8 showcases detected windshields from down-moving vehicles, demonstrating the accuracy of windshield segmentation.

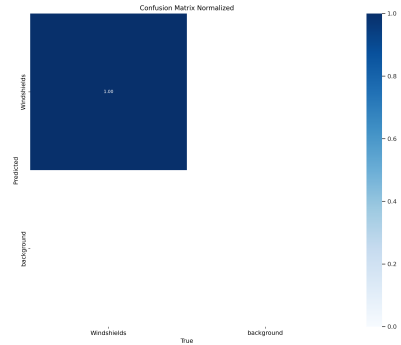


Fig. 6: Windshield detection - Normalized Confusion Matrix

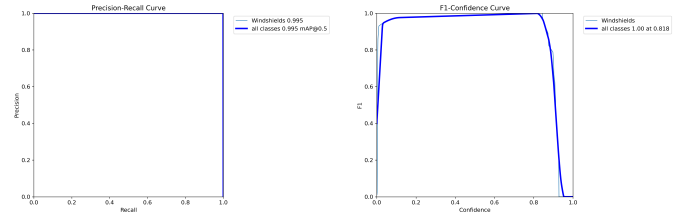


Fig. 7: Windshield Detection - Precision-Recall and F1 Curve



Fig. 8: Examples of Detected Windshields

Following windshield segmentation, the passenger classification model was applied to determine whether an occupant was present in the passenger seat. The performance of passenger classification model is given in Table V, and it confirms the ability to flawlessly identify passengers.

TABLE V: Passenger Classification Performance of YOLOv8n-cls Model

Classes	top1_acc	top5_acc
all	0.99	1.00

The confusion matrix displaying the performance of YOLOv8n-cls in classifying passenger presence within vehicles is shown in Fig. 9.

Fig. 10(a) represents an example of a detected windshield with a passenger and 10(b) represents windshield without a passenger.

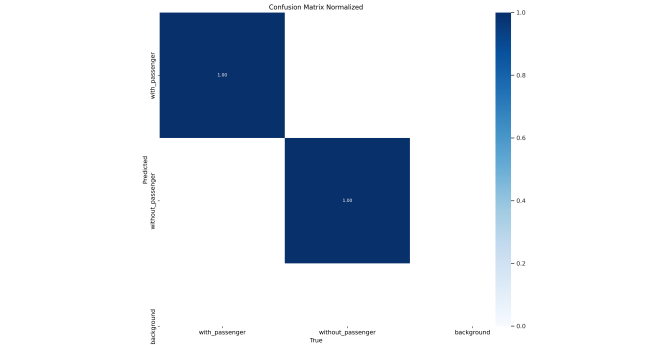


Fig. 9: Passenger Classification - Normalized Confusion Matrix

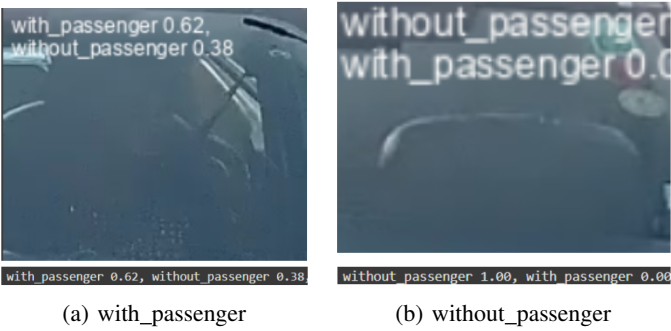


Fig. 10: Passenger Classification Results

• Seat Belt Detection

The seat belt detection model classified images into two categories: “with seatbelt” and “without seatbelt”. The model achieved 87.5% precision and an F1-score of 85.9%, indicating high reliability. Additionally, CLAHE preprocessing significantly improved detection accuracy from 82% to 88%, especially in cases where seat belts were difficult to distinguish due to low contrast or similar-coloured clothing.

If the model’s confidence score fell below 0.25, the image underwent CLAHE enhancement before final prediction. This enhancement significantly improved visibility and reduced false negatives.

TABLE VI: Seat Belt Detection Performance of YOLOv8n Model

Model	P	R	mAP@50	mAP@50-95
all	0.933	0.797	0.875	0.472
with_seatbelt	0.906	0.761	0.912	0.522
without_seatbelt	0.961	0.833	0.838	0.423

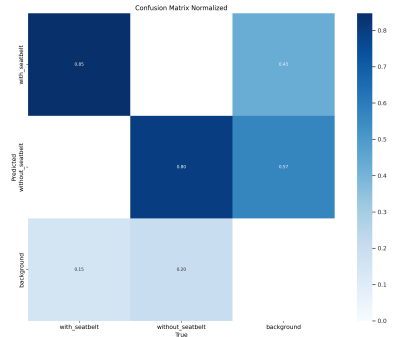


Fig. 11: Seatbelt Detection - Normalized Confusion Matrix

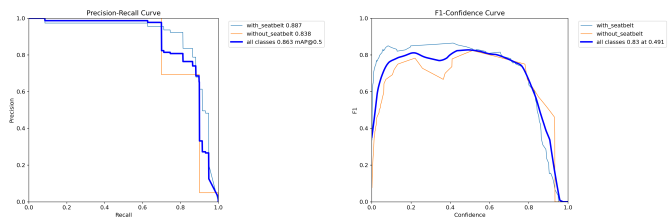


Fig. 12: Seatbelt Detection - Precision-Recall and F1 Curve

The following images show accurate detection of seat belt usage using the proposed YOLOv8n model [Fig. 13], and a case where the model fails to detect a seat belt due to the low contrast between clothing and the seat belt [Fig. 14].

The same image from Fig. 14 after CLAHE pre-processing and successfully improving seat belt visibility and correcting the mis-classification is illustrated in Fig. 15.

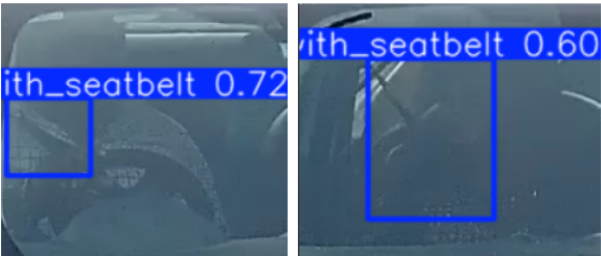


Fig. 13: Successful Seat Belt Detection Results

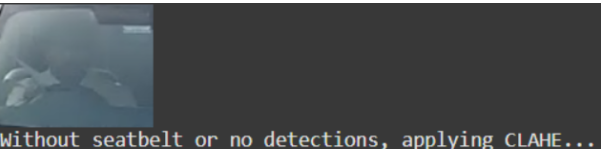


Fig. 14: Example of Misclassified Seat Belt Detection

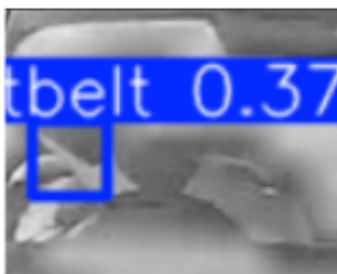


Fig. 15: Seatbelt Detection After CLAHE Enhancement

V. CONCLUSION

This paper presents an automated seat belt detection framework leveraging YOLOv8 models and CLAHE enhancement to improve classification accuracy. The proposed system effectively detects four-wheelers, classifies their direction, isolates windshield regions, and determines seat belt usage with high reliability. Experimental results confirm that YOLOv8n is well-suited for real-time traffic enforcement applications, balancing accuracy and speed. Future work will focus on expanding the dataset to include more diverse traffic and light conditions, integrating additional occlusion handling techniques, and exploring hybrid approaches combining YOLO with transformer-based architectures to further enhance model robustness.

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